



# Ansys Fluent Simulation Report

|                |                   |
|----------------|-------------------|
| <b>Analyst</b> | ShubhamR          |
| <b>Date</b>    | 2/8/2026 11:17 AM |

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# System Information

|                 |                                                                |
|-----------------|----------------------------------------------------------------|
| Application     | Fluent                                                         |
| Settings        | 2d, double precision, density-based explicit, Spalart-Allmaras |
| Version         | 25.2.0-10204                                                   |
| Source Revision | 5eecd5d865                                                     |
| Build Time      | Jun 16 2025 10:44:41 EDT                                       |
| CPU             | AMD Ryzen 7 6800H with Radeon Graphics                         |
| OS              | Windows                                                        |

# Geometry and Mesh

## Mesh Size

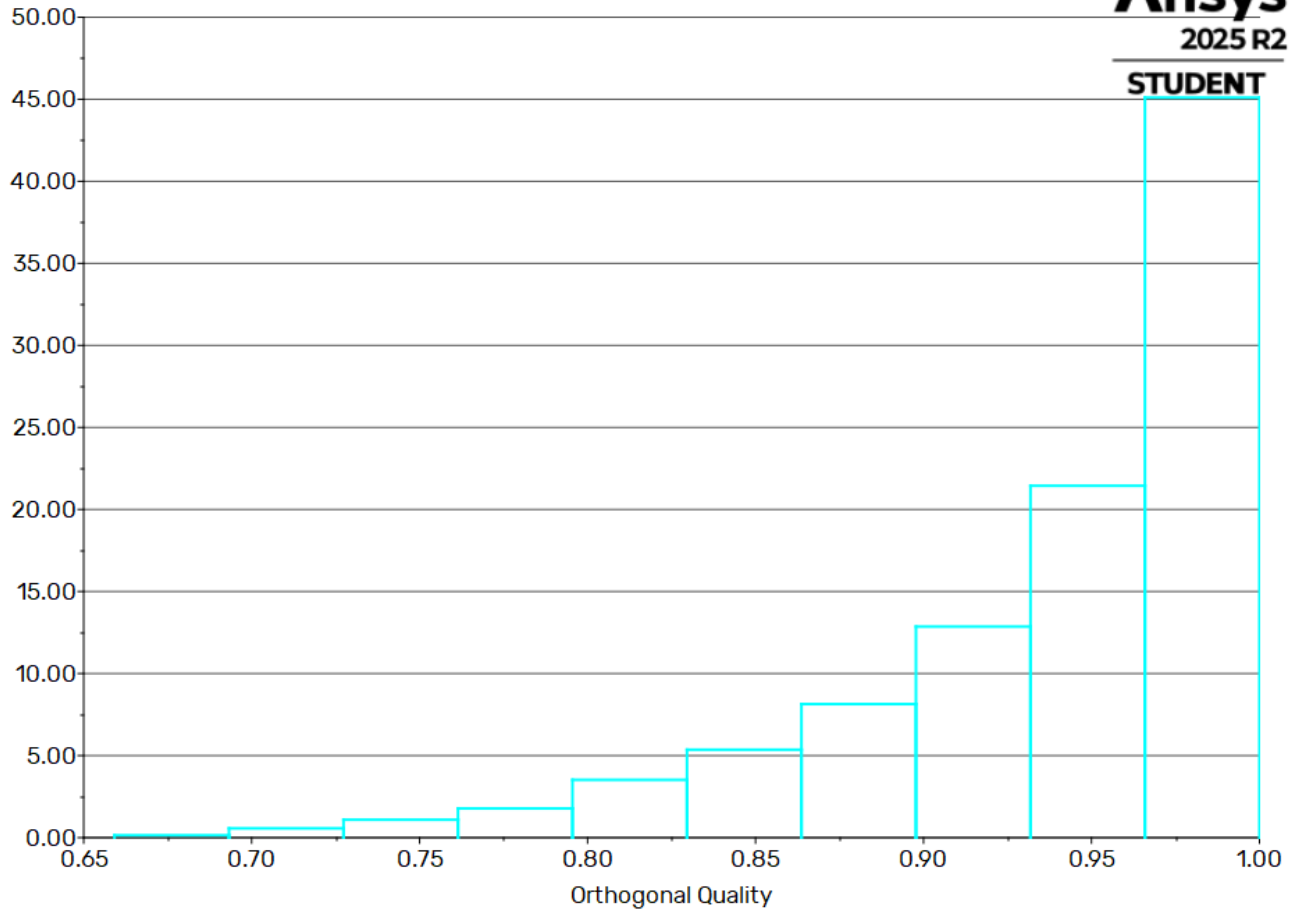
| Cells | Faces | Nodes |
|-------|-------|-------|
| 39    | 140   | 0     |
| 118   | 415   | 0     |
| 351   | 1192  | 0     |
| 1023  | 3254  | 0     |
| 3288  | 9800  | 0     |
| 22914 | 35063 | 12153 |

## Mesh Quality

| Name  | Type     | Min Orthogonal Quality | Max Aspect Ratio |
|-------|----------|------------------------|------------------|
| fluid | Tri Cell | 0.65988415             | 3.7977613        |
| solid | Tri Cell | 0.65917937             | 3.7967206        |

## Orthogonal Quality

Histogram of Orthogonal Quality



# Simulation Setup

## Physics

### Models

| Model             | Settings                          |
|-------------------|-----------------------------------|
| Space             | 2D                                |
| Time              | Steady                            |
| Viscous           | Spalart-Allmaras turbulence model |
| Production Option | Vorticity                         |
| Heat Transfer     | Enabled                           |

### Material Properties

|                      |                     |
|----------------------|---------------------|
| – Fluid              |                     |
| – air                |                     |
| Density              | ideal gas           |
| Cp (Specific Heat)   | 1006.43 J/(kg K)    |
| Thermal Conductivity | 0.0242 W/(m K)      |
| Viscosity            | 1.7894e-05 kg/(m s) |
| Molecular Weight     | 28.966 kg/kmol      |
| – Solid              |                     |
| – aluminum           |                     |
| Density              | 2719 kg/m^3         |
| Cp (Specific Heat)   | 871 J/(kg K)        |
| Thermal Conductivity | 202.4 W/(m K)       |

### Cell Zone Conditions

|         |  |
|---------|--|
| – Fluid |  |
| – fluid |  |
|         |  |

|                       |          |
|-----------------------|----------|
| Material Name         | air      |
| Specify source terms? | no       |
| Specify fixed values? | no       |
| Frame Motion?         | no       |
| Laminar zone?         | no       |
| Porous zone?          | no       |
| – Solid               |          |
| – solid               |          |
| Material Name         | aluminum |
| Specify source terms? | no       |
| Frame Motion?         | no       |
| Solid Motion?         | no       |

## Boundary Conditions

|                                   |                           |
|-----------------------------------|---------------------------|
| – Inlet                           |                           |
| – inlet                           |                           |
| Gauge Pressure [Pa]               | 101325                    |
| Mach Number                       | 1.1                       |
| Temperature [K]                   | 300                       |
| Component of Flow Direction (x,y) | (1, 0)                    |
| Turbulence Specification Method   | Turbulent Viscosity Ratio |
| Turbulent Viscosity Ratio         | 10                        |
| – outlet                          |                           |
| Gauge Pressure [Pa]               | 101325                    |
| Mach Number                       | 1.1                       |
| Temperature [K]                   | 300                       |
| Component of Flow Direction (x,y) | (1, 0)                    |
| Turbulence Specification Method   | Turbulent Viscosity Ratio |
| Turbulent Viscosity Ratio         | 10                        |
| – Wall                            |                           |
| – soild_wall                      |                           |

|                                |                 |
|--------------------------------|-----------------|
| Wall Thickness [m]             | 0               |
| Heat Generation Rate [W/m^3]   | 0               |
| Material Name                  | aluminum        |
| Thermal BC Type                | Heat Flux       |
| Heat Flux [W/m^2]              | 0               |
| Convective Augmentation Factor | 1               |
| – soild_wall:001               |                 |
| Wall Thickness [m]             | 0               |
| Heat Generation Rate [W/m^3]   | 0               |
| Material Name                  | aluminum        |
| Thermal BC Type                | Coupled         |
| Wall Motion                    | Stationary Wall |
| Shear Boundary Condition       | No Slip         |
| Wall Surface Roughness         | Standard        |
| Wall Roughness Height [m]      | 0               |
| Wall Roughness Constant        | 0.5             |
| Convective Augmentation Factor | 1               |
| – soild_wall:001-shadow        |                 |
| Wall Thickness [m]             | 0               |
| Heat Generation Rate [W/m^3]   | 0               |
| Material Name                  | aluminum        |
| Thermal BC Type                | Coupled         |
| Convective Augmentation Factor | 1               |

## Reference Values

|                    |              |
|--------------------|--------------|
| <b>Area</b>        | 1 m^2        |
| <b>Density</b>     | 1.225 kg/m^3 |
| <b>Depth</b>       | 1 m          |
| <b>Enthalpy</b>    | 0 J/kg       |
| <b>Length</b>      | 1 m          |
| <b>Pressure</b>    | 0 Pa         |
| <b>Temperature</b> | 288.16 K     |

|                                   |                     |
|-----------------------------------|---------------------|
| <b>Velocity</b>                   | 1 m/s               |
| <b>Viscosity</b>                  | 1.7894e-05 kg/(m s) |
| <b>Ratio of Specific Heats</b>    | 1.4                 |
| <b>Yplus for Heat Tran. Coef.</b> | 300                 |
| <b>Reference Zone</b>             | fluid               |

## Solver Settings

|                               |                     |
|-------------------------------|---------------------|
| – Equations                   |                     |
| Flow                          | True                |
| Modified Turbulent Viscosity  | True                |
| – Numerics                    |                     |
| Absolute Velocity Formulation | True                |
| – Under-Relaxation Factors    |                     |
| Modified Turbulent Viscosity  | 0.8                 |
| Turbulent Viscosity           | 1                   |
| Solid                         | 1                   |
| – Discretization Scheme       |                     |
| Flow                          | Second Order Upwind |
| Modified Turbulent Viscosity  | Second Order Upwind |
| – Time Marching               |                     |
| Solver                        | Explicit            |
| Courant Number                | 0.05                |
| Residual Smoothing Iterations | 0 (disabled)        |
| Residual Smoothing Factor     | 0.5                 |
| – Multigrid Solver            |                     |
| Coarse Grid Levels            | 5                   |
| Fine Grid Iterations          | 1                   |
| Final Iterations              | 0                   |
| Time Step Reduction           | 0.9                 |
| Correction Reduction          | 0.6                 |
| Species Correction Reduction  | 1                   |



|                                |        |
|--------------------------------|--------|
| Correction Smoothing Factor    | 0.5    |
| – Multi-Stage Scheme           |        |
| 1                              | 0.2075 |
| 2                              | 0.5915 |
| 3                              | 1      |
| – Solution Limits              |        |
| Minimum Absolute Pressure [Pa] | 1      |
| Maximum Absolute Pressure [Pa] | 5e+10  |
| Minimum Static Temperature [K] | 1      |
| Maximum Static Temperature [K] | 5000   |
| Maximum Turb. Viscosity Ratio  | 100000 |

## Run Information

|                               |                 |
|-------------------------------|-----------------|
| <b>Number of Machines</b>     | 1               |
| <b>Number of Cores</b>        | 2               |
| <b>Case Read</b>              | 7.168 seconds   |
| <b>Iteration</b>              | 153.359 seconds |
| <b>AMG</b>                    | 1.133 seconds   |
| <b>Virtual Current Memory</b> | 1.0861 GB       |
| <b>Virtual Peak Memory</b>    | 1.1983 GB       |
| <b>Memory Per M Cell</b>      | 19.0502         |

# Solution Status

Iterations: 4585

|            | Value        | Absolute Criteria | Convergence Status |
|------------|--------------|-------------------|--------------------|
| continuity | 0.0009998756 | 0.001             | Converged          |
| x-velocity | 0.0001775964 | 0.001             | Converged          |
| y-velocity | 0.0005544375 | 0.001             | Converged          |
| energy     | 0.0008708995 | 0.001             | Converged          |
| nut        | 0.0002517476 | 0.001             | Converged          |

# Named Expressions

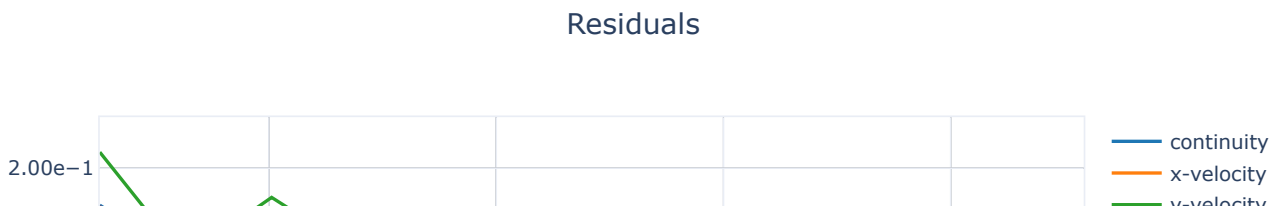
| Expression | Definition                     | Value     | Unit | Used In                             | Description |
|------------|--------------------------------|-----------|------|-------------------------------------|-------------|
| expr1      | Maximum(MachNumber, ['fluid']) | 1.9877734 |      | report-def-0-sa (Report Definition) |             |

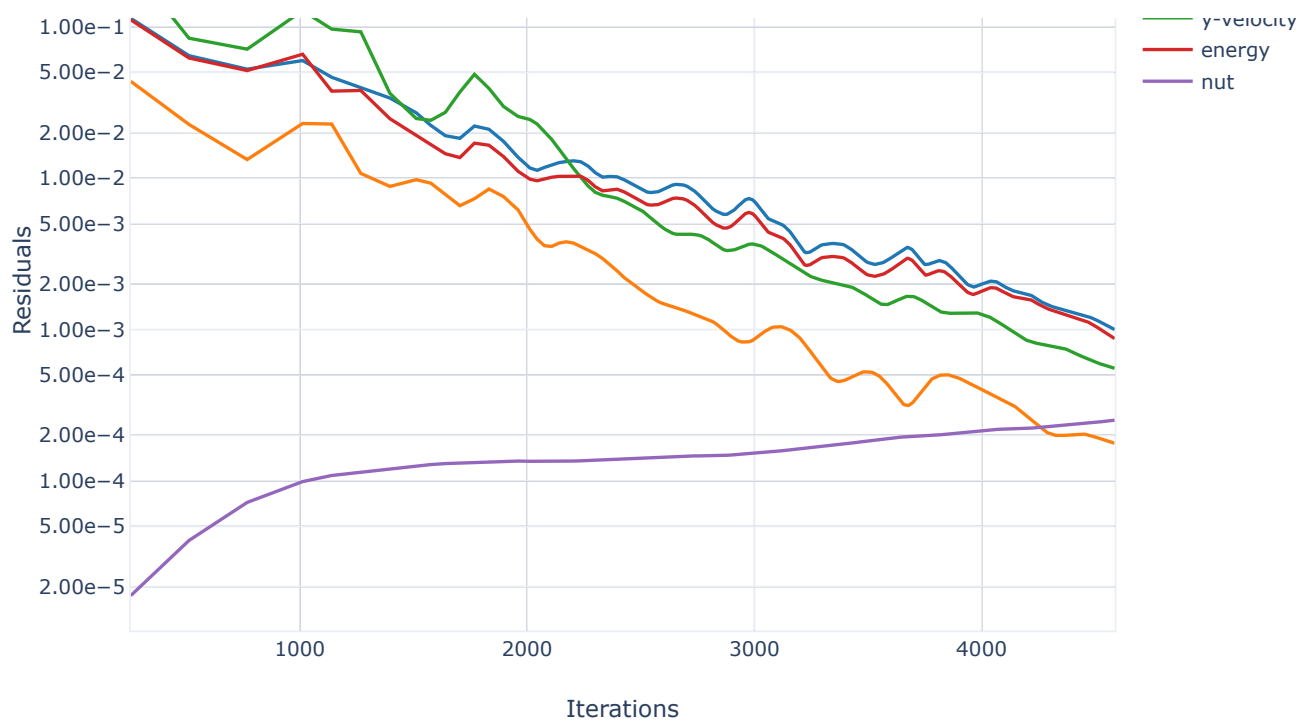
# Report Definitions

|                 |          |  |
|-----------------|----------|--|
| report-def-0-sa | 1.987773 |  |
|-----------------|----------|--|

# Plots

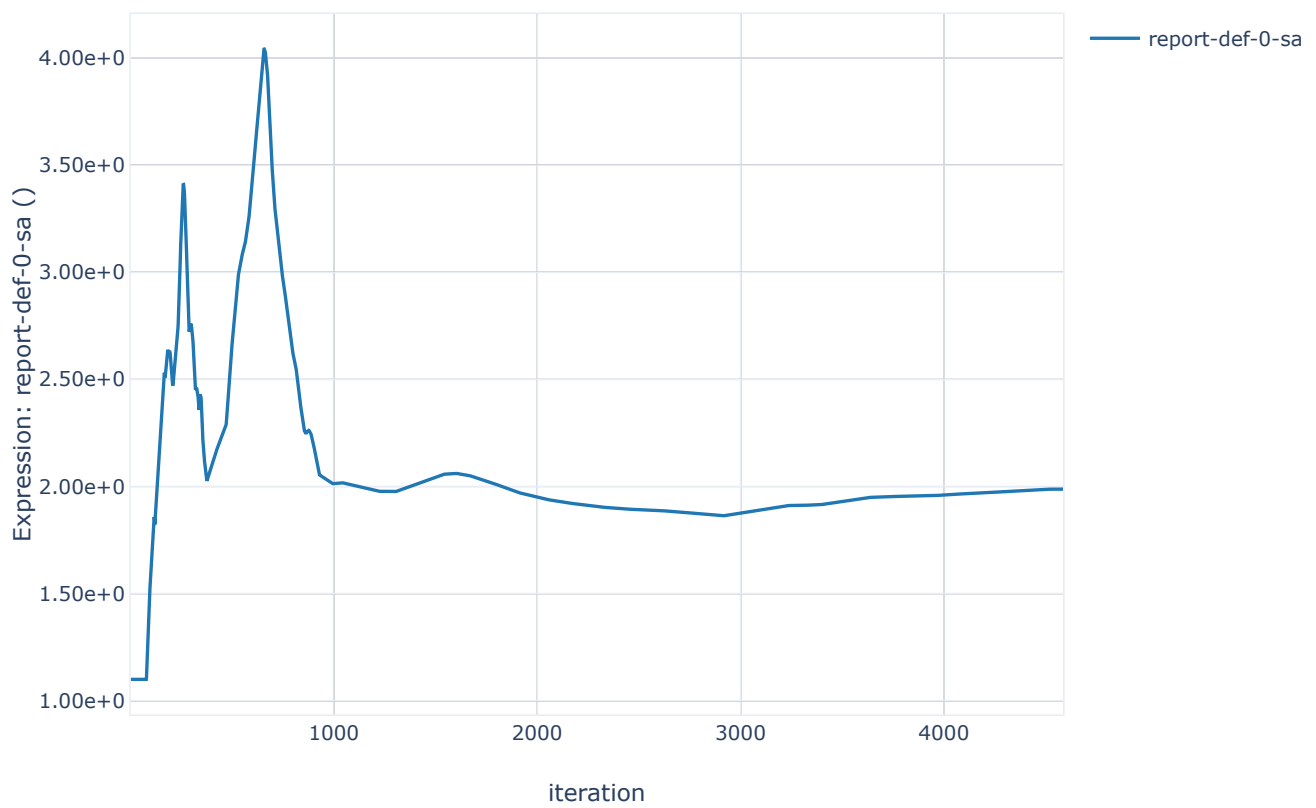
# Residuals





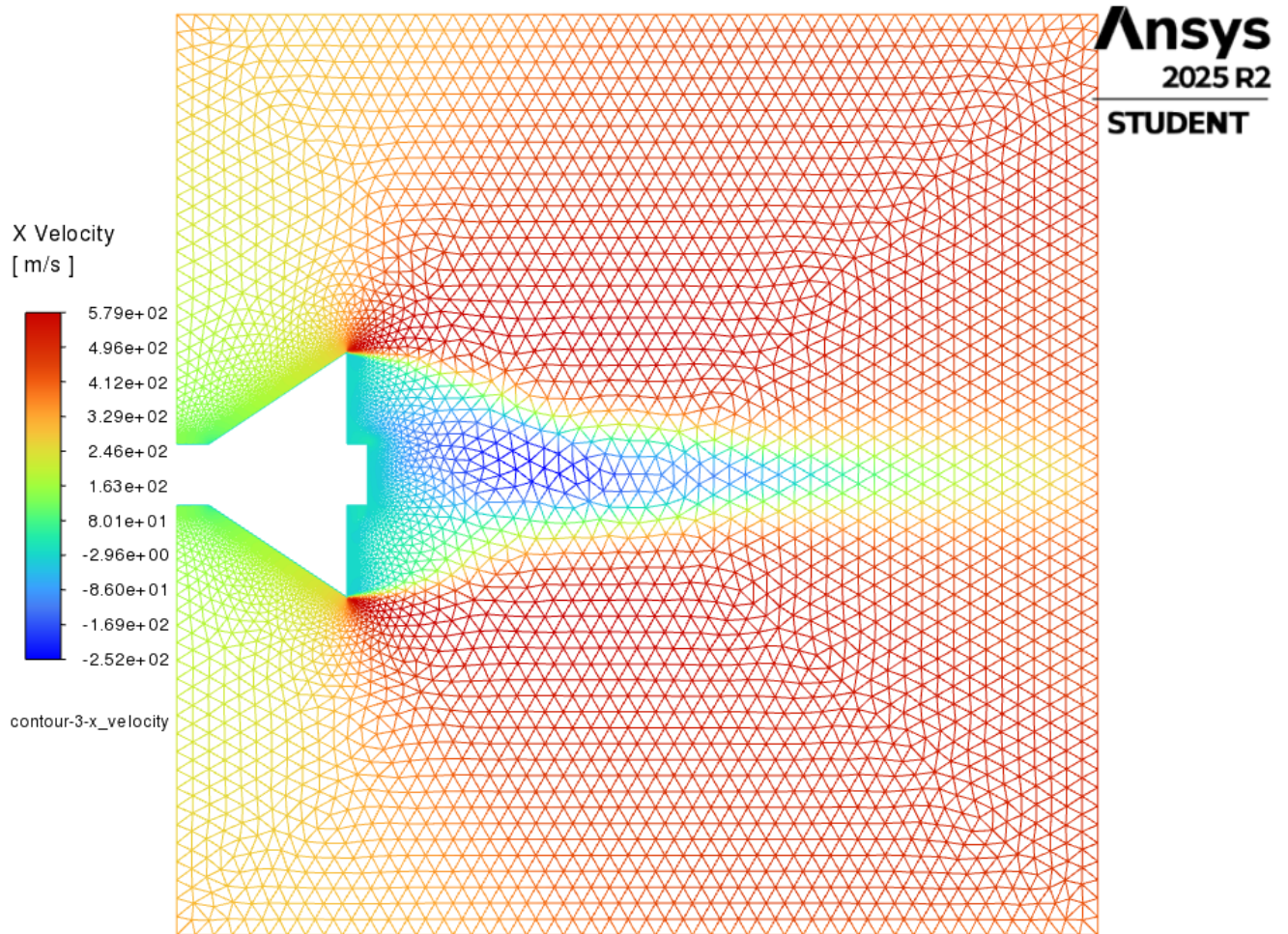
report-def-0-rplot

report-def-0-rplot



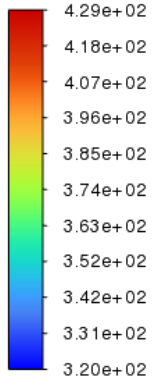
Contours

contour-3-x\_velocity

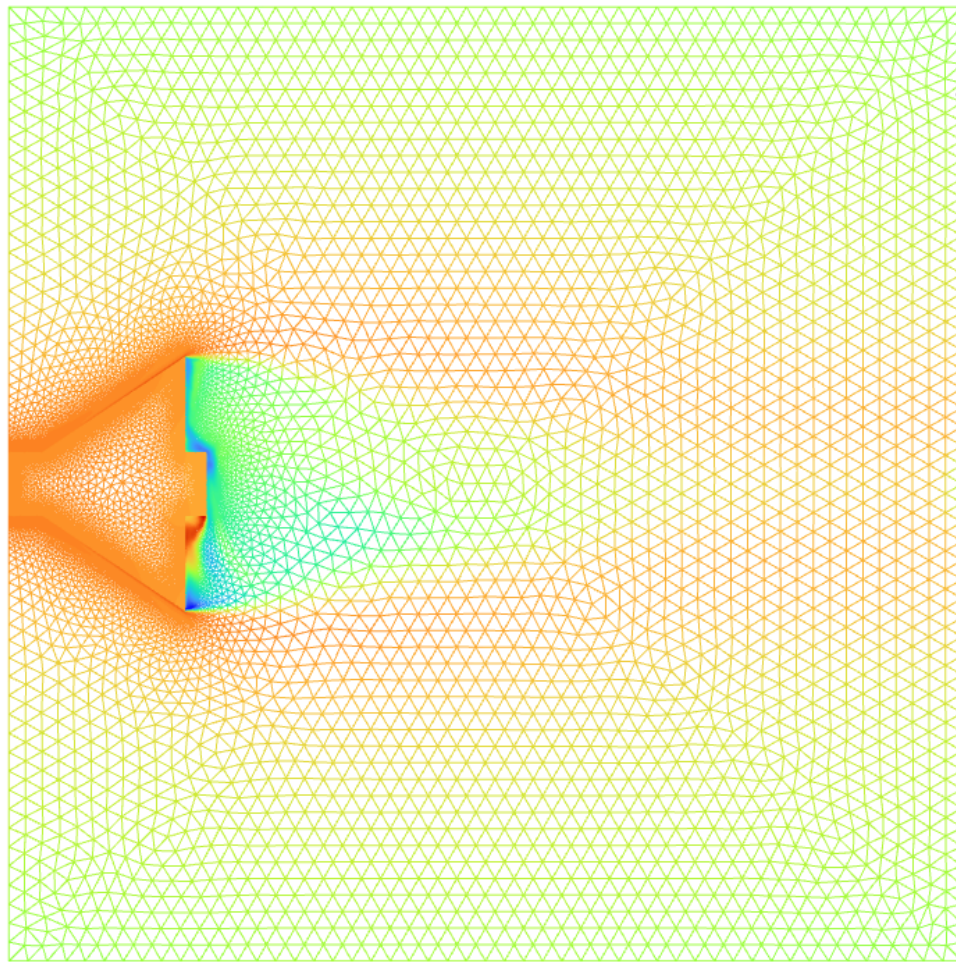


contour-2-total\_temperature

Total Temperature  
[ K ]



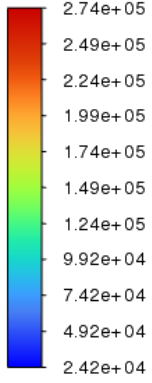
contour-2-total\_tem-  
perature



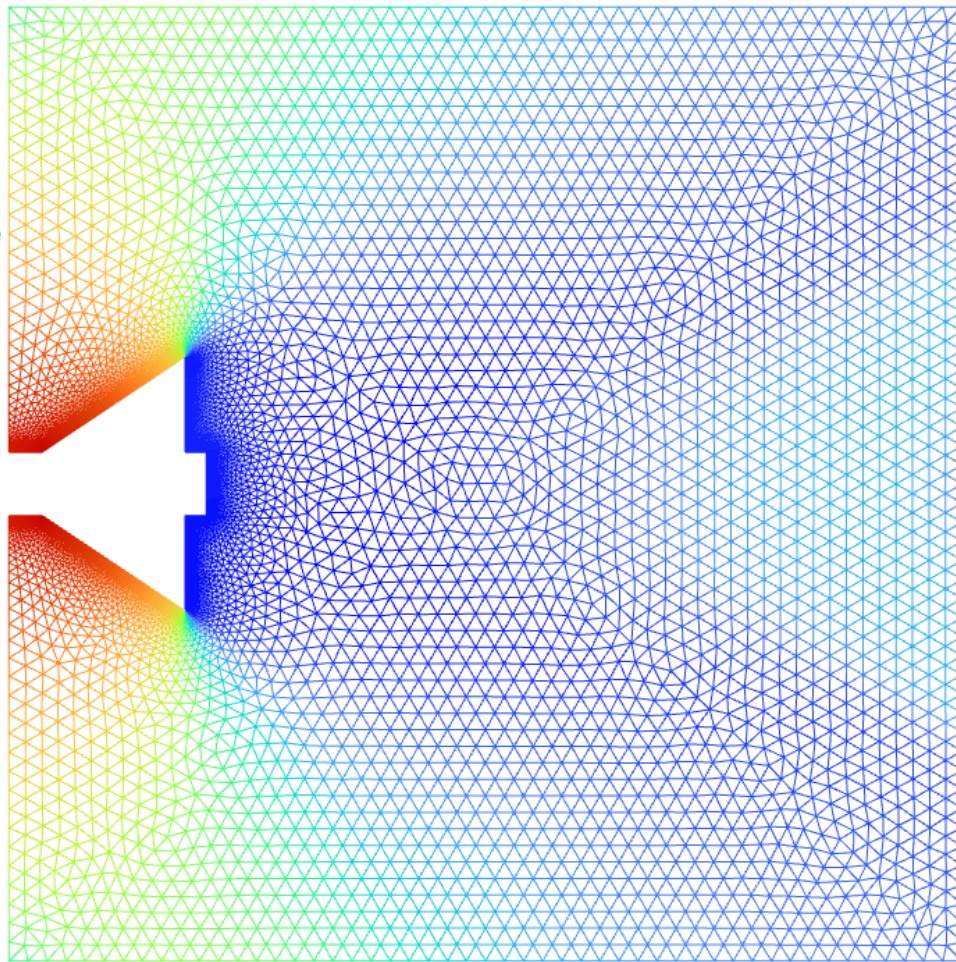
contour-1-static\_pressure



Static Pressure  
[ Pa ]



contour-1-static\_pres-  
-sure

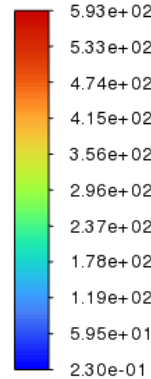


# Vectors

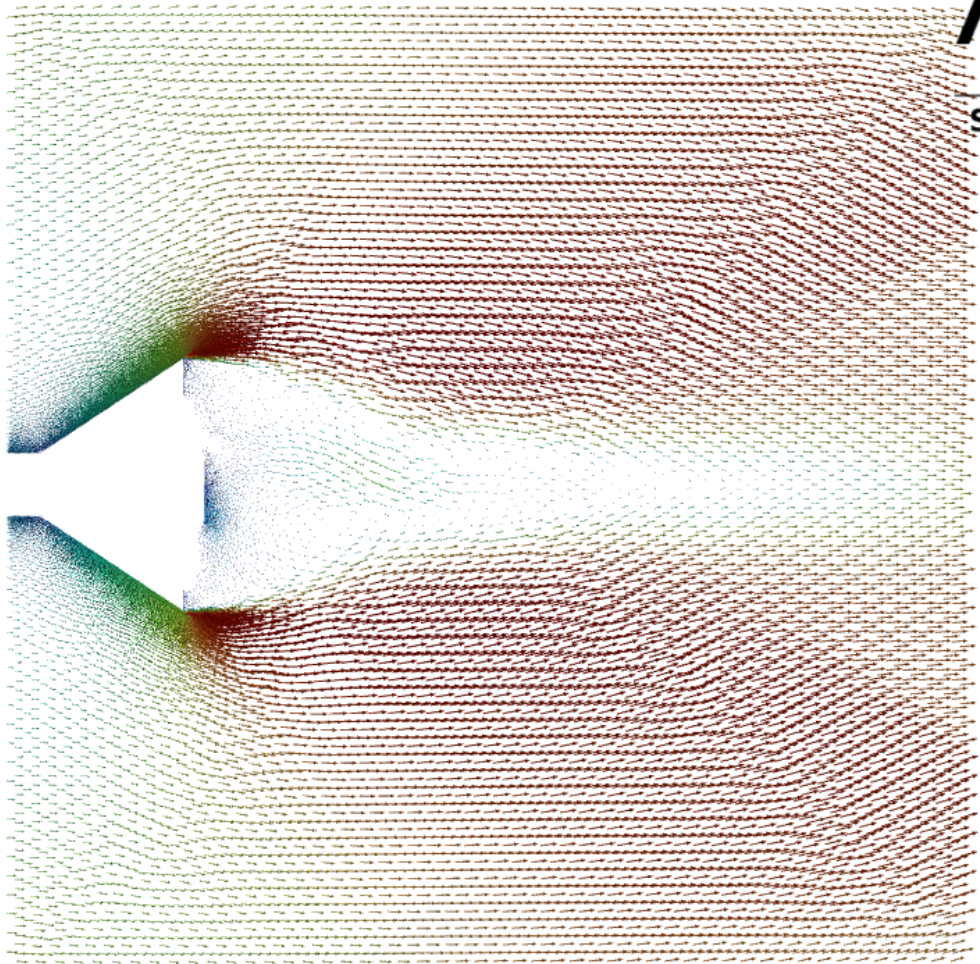
vector-1

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Velocity Magnitude  
[ m/s ]

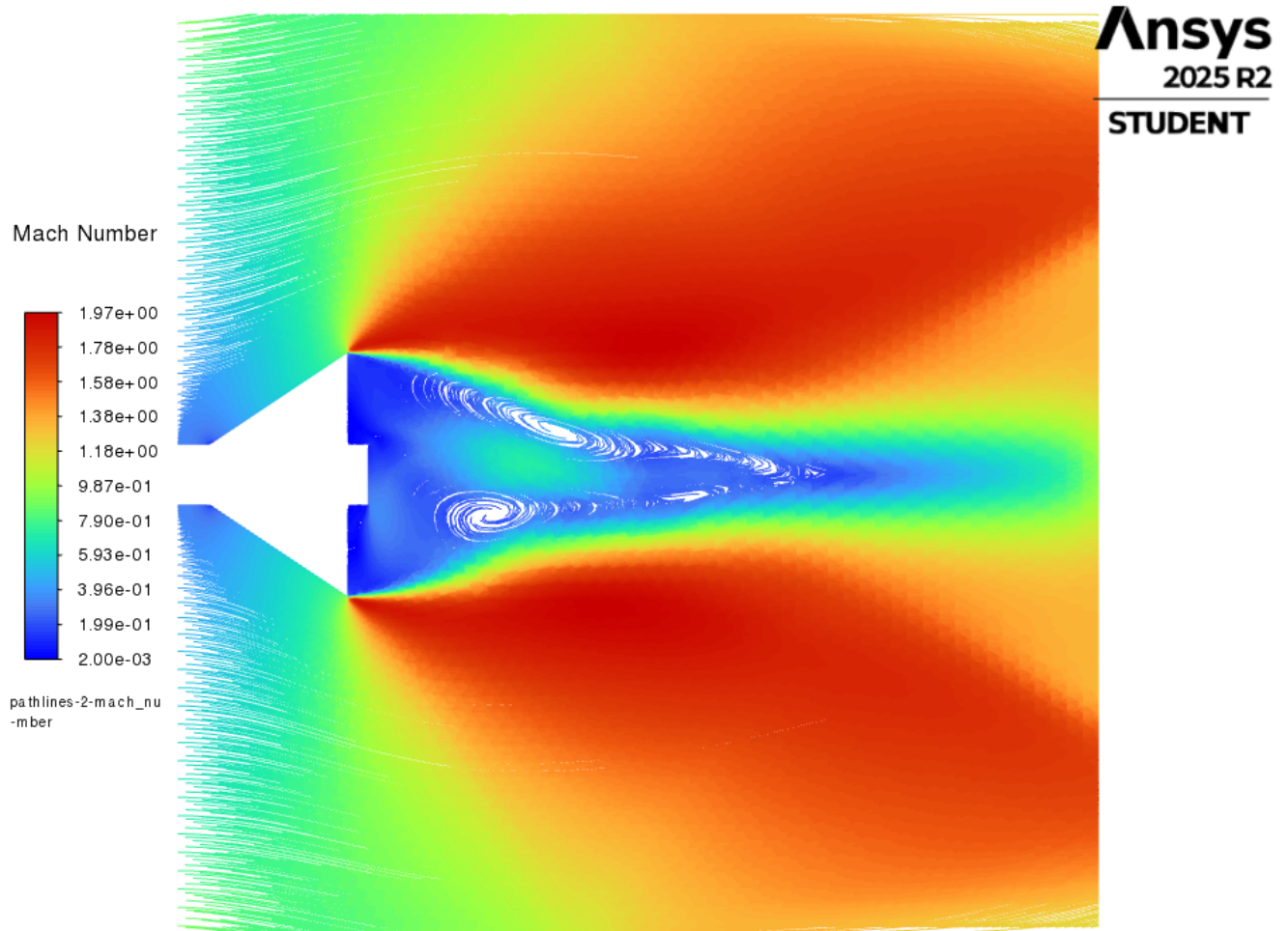


vector-1



## Pathlines

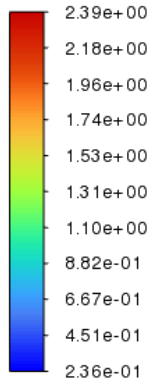
pathlines-2-mach\_number



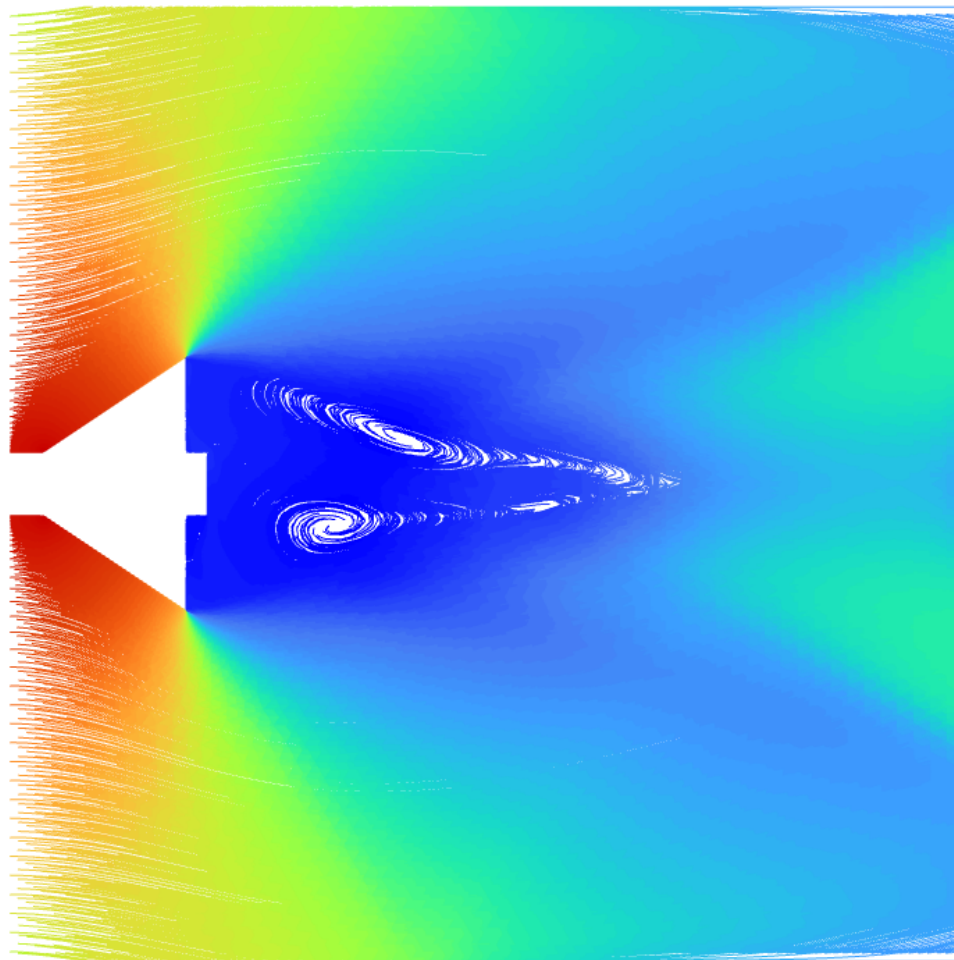
pathlines-1-density



Density  
[ kg/m<sup>3</sup> ]

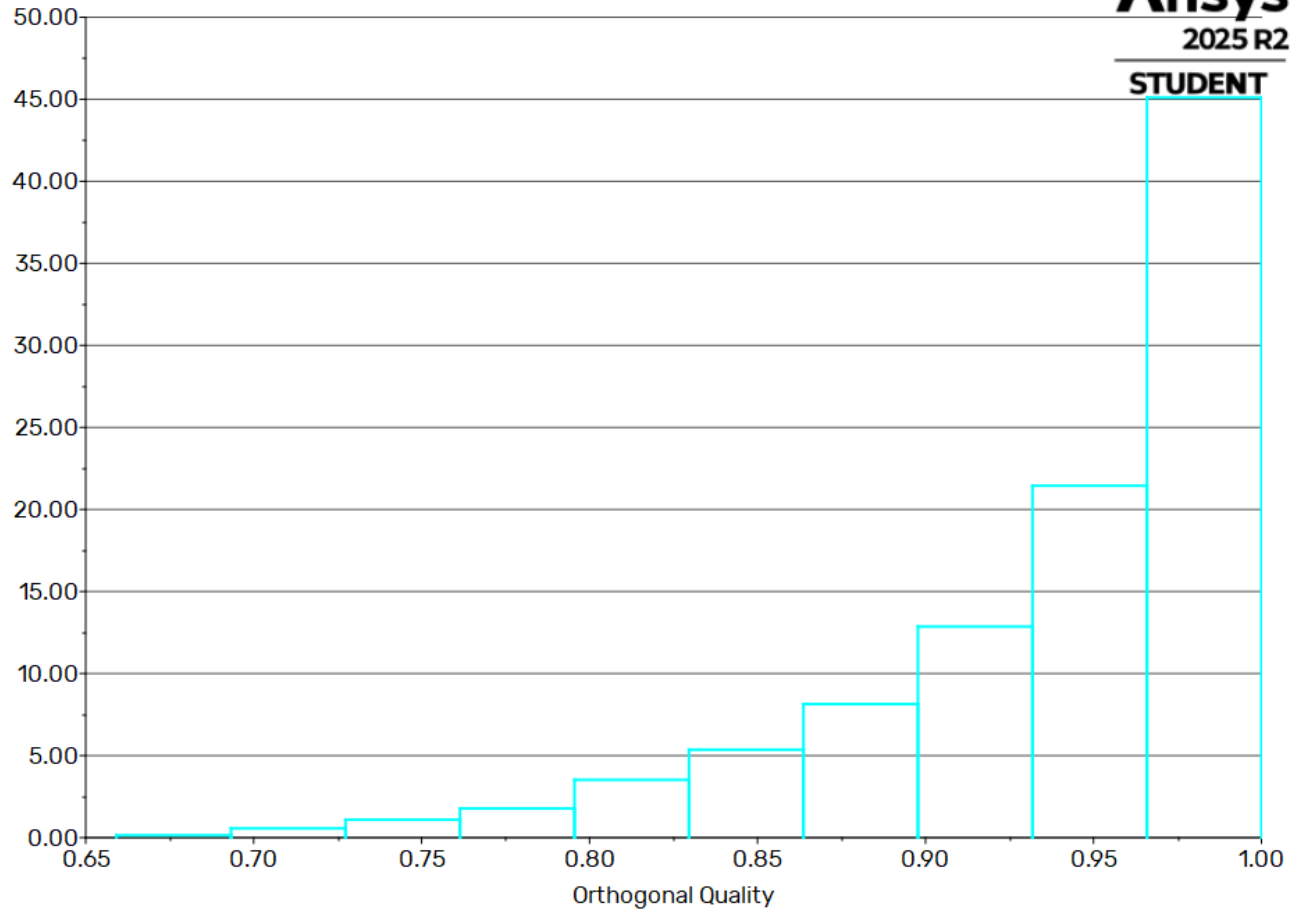


pathlines-1-density

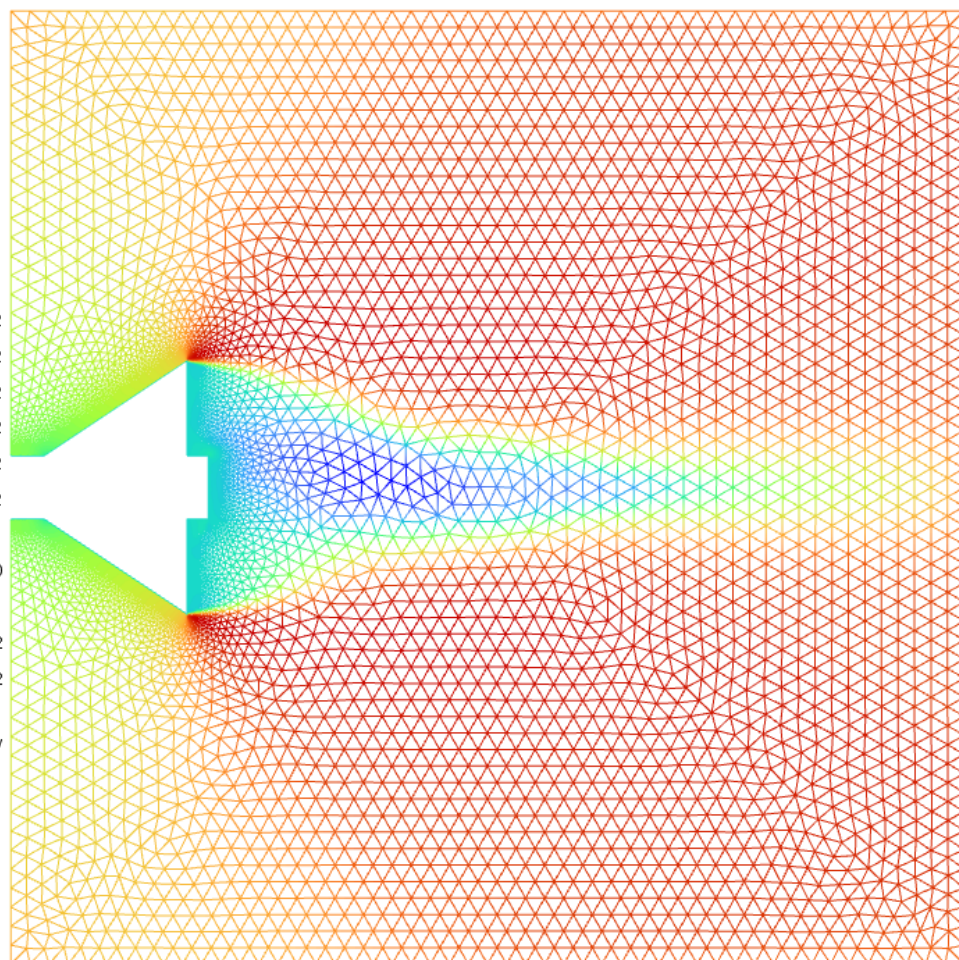
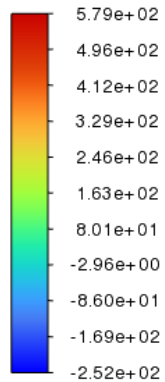


**ANSYS**  
**2025 R2**

Histogram of Orthogonal Quality

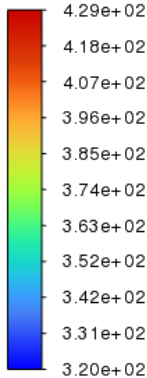


X Velocity  
[ m/s ]

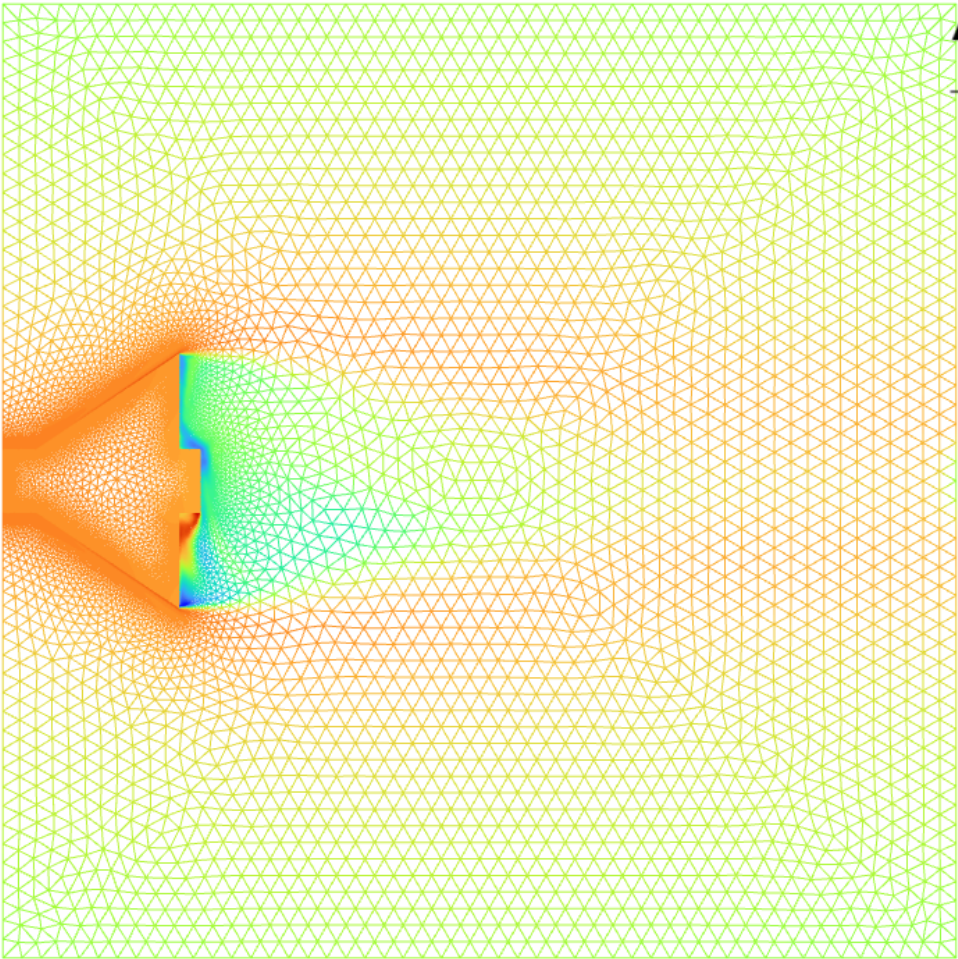




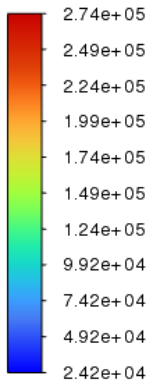
Total Temperature  
[ K ]



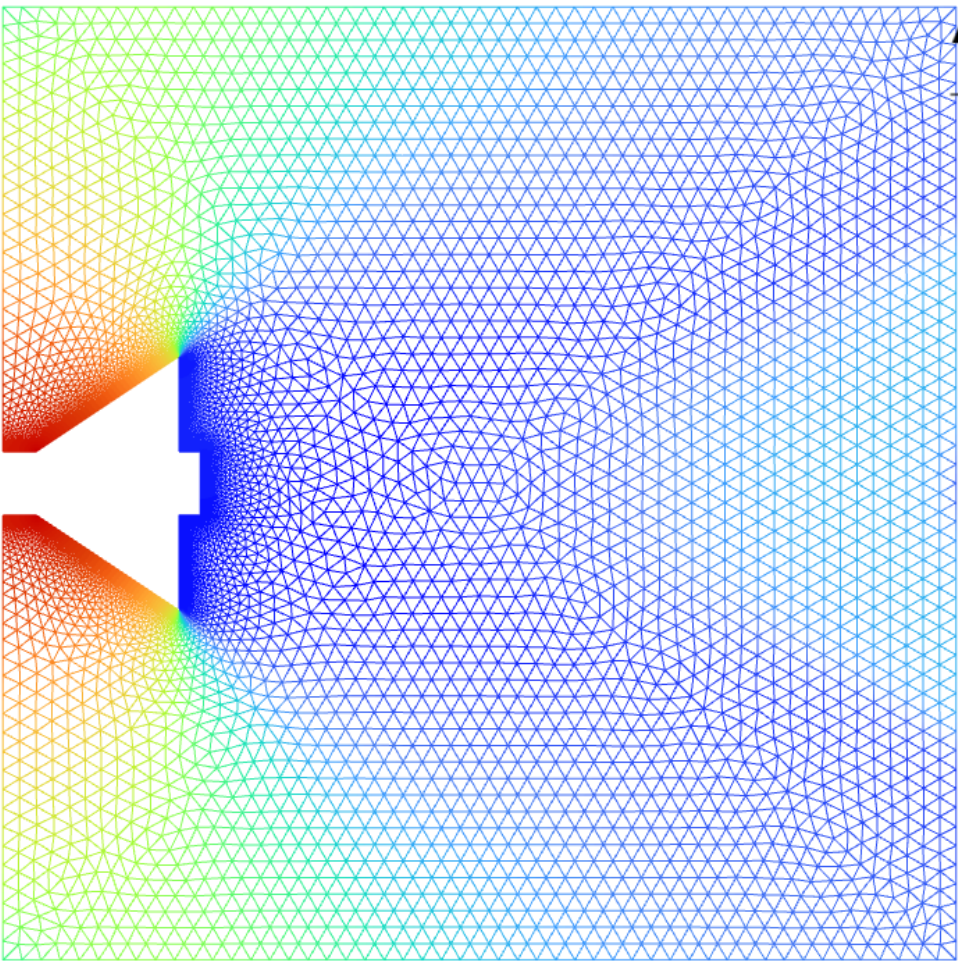
contour-2-total\_tem-  
perature



Static Pressure  
[ Pa ]

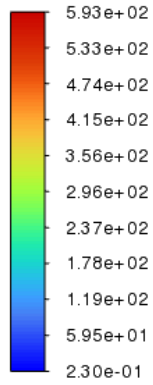


contour-1-static\_pres-  
sure

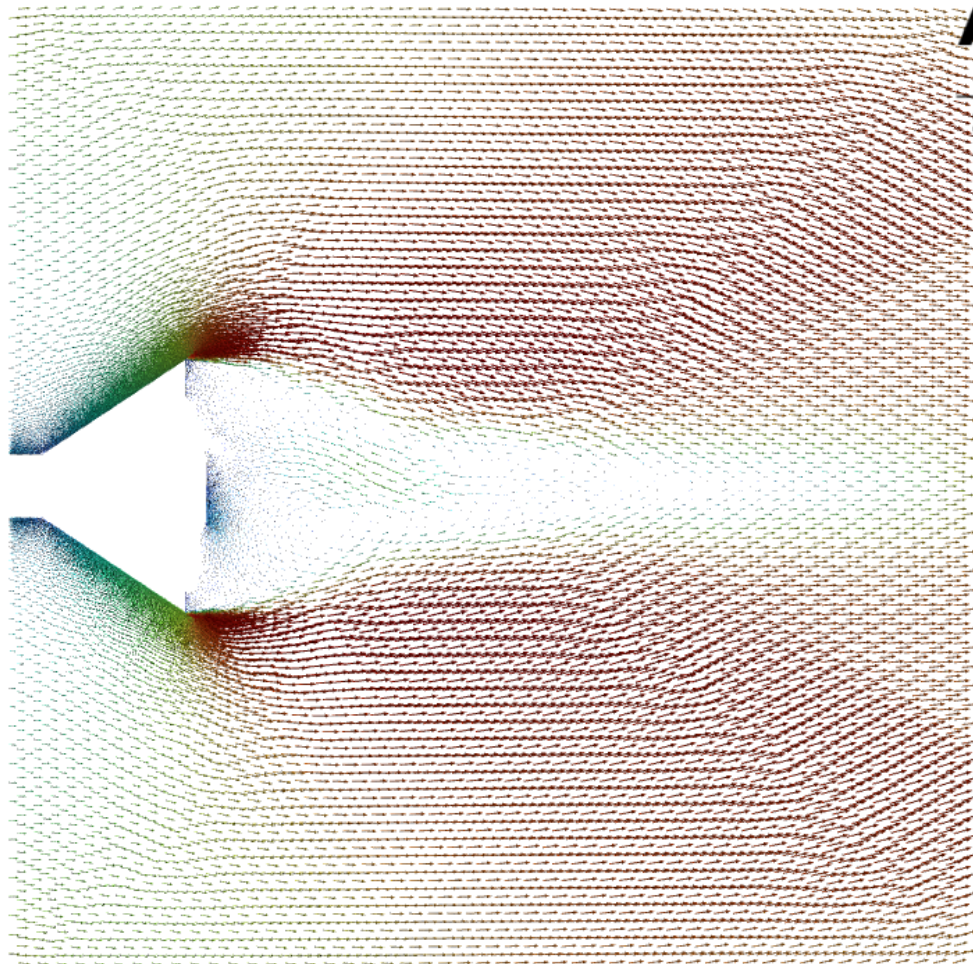




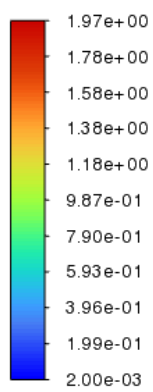
Velocity Magnitude  
[ m/s ]



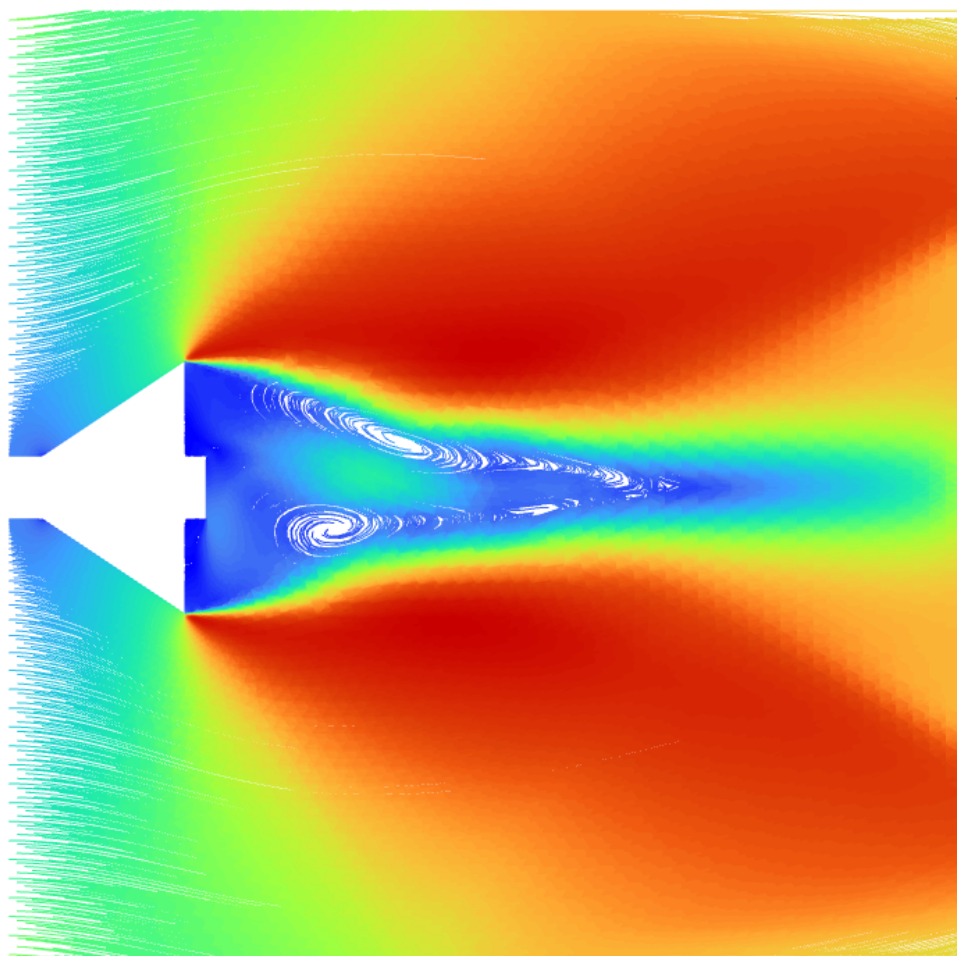
vector-1



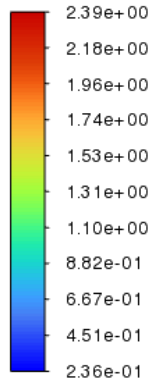
Mach Number



pathlines-2-mach\_number



Density  
[ kg/m<sup>3</sup> ]



pathlines-1-density

